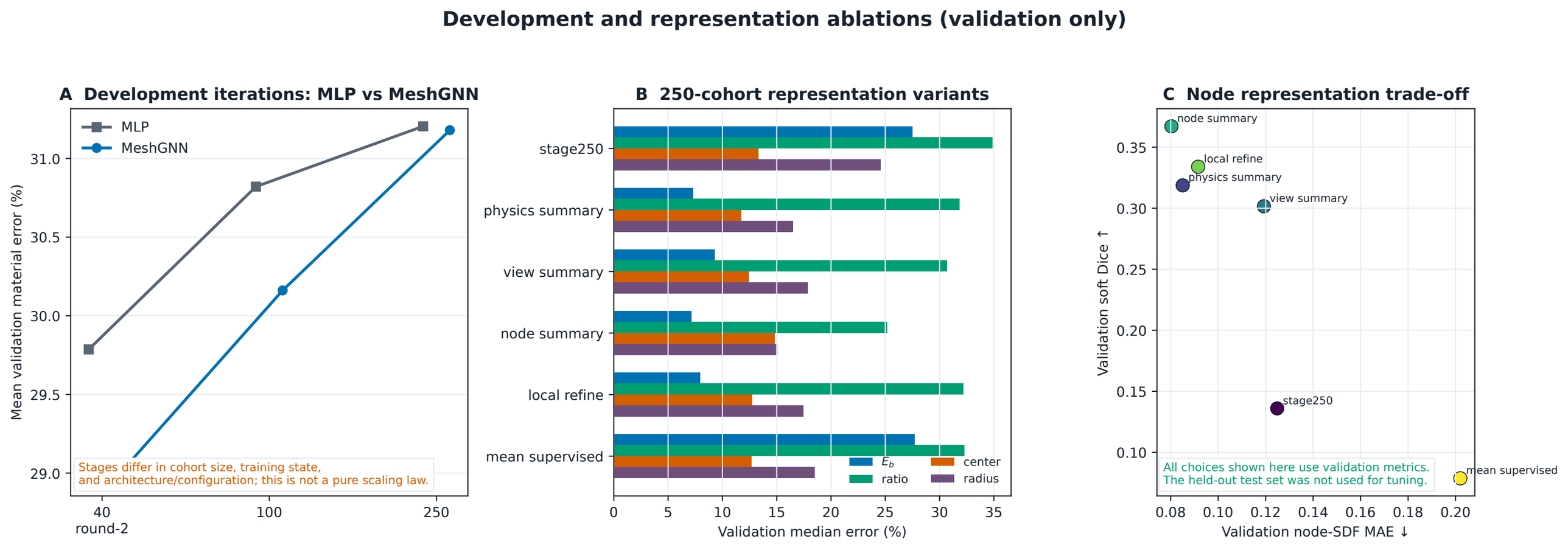
Development and representation ablations (validation only)
A Development iterations: MLP vs MeshGNN
MLP
MeshGNN
Mean validation material error (%)
31.0
30.5
30.0
29.5
29.0
Stages differ in cohort size, training state,
and architecture/configuration; this is not a pure scaling law.
40
round-2
100
250
B 250-cohort representation variants
stage250
physics summary
view summary
node summary
local refine
mean supervised
E_b
ratio
center
radius
0
5
10
15
20
25
30
35
Validation median error (%)
C Node representation trade-off
node summary
local refine
physics summary
view summary
stage250
mean supervised
Validation soft Dice ↑
0.35
0.30
0.25
0.20
0.15
0.10
All choices shown here use validation metrics.
The held-out test set was not used for tuning.
0.08
0.10
0.12
0.14
0.16
0.18
0.20
Validation node-SDF MAE ↓